106 Stochastic Processes

MTL[redacted]: Probability and [redacted]

(Lecture [redacted]) 9

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Generating Functions: We now consider some functions that generate probabilities or moments of a random variable. The simplest type of generating function in probability theory is the one associated with integer-valued random variables. Let X be a random variable, and let

$$p_k = P(X = k), \;\; k = 0, 1, 2, ...$$

with $\sum_{k=0}^{\infty} p_k = 1$.

Probability Generating Function: The function defined by

$$P(s) = \sum_{k=0}^{\infty} p_k s^k, \tag{1}$$

which surely converges for $|s| \leq 1$, is called the *probability generating function* (PGF) of X.

Example 1. Let X be a random variable with PMF

$$P(X = k) = pq^k, \;\; k = 0, 1, 2, ...; \;\; 0 < p < 1, \;\; q = 1 - p.$$

Then the PGF is given by

$$P(s) = \sum_{k=0}^{\infty} s^k pq^k = \frac{p}{1 - sq}, \;\; |s| \leq 1.$$

Remark 1. Since $P(1) = 1$, series (1) is uniformly and absolutely convergent in $|s| \leq 1$ and the PGF P is a continuous function of s. It determines the PGF uniquely, since $P(s)$ can be represented in a unique manner as a power series.

Remark 2. Since a power series with radius of convergence r can be differentiated with termwise any number of times in $(-r, r)$, it follows that

$$P^{(k)}(s) = \sum_{n=k}^{\infty} n(n-1)(n-2)...(n-k+1)P(X = n)s^{n-k},$$

where $P^{(k)}$ is the k^{th} derivative of P. The series converges at least for $-1 < s < 1$. For $s = 1$ the right side reduces formally to $E[X(X-1)...(X-k+1)]$, which is the k^{th} *factorial moment* of

X whenever it exists. In particular, if $E(X) < \infty$, then $P'(1) = E(X)$, and if $E(X^2) < \infty$, then $P''(1) = E(X(X-1))$ and $Var(X) = E(X^2) - (E(X))^2 = P''(1) - [P'(1)]^2 + P'(1)$.

Example 2. Consider a random variable with PMF

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \;\; k = 0, 1, 2, ...$$

The PGF of X is given by

$$P(s) = \sum_{k=0}^{\infty} s^k \left(\frac{e^{-\lambda}\lambda^k}{k!} \right) = e^{-\lambda(1-s)} \;\; \forall \;\; s.$$

We have

$$P'(s) = \lambda e^{-\lambda(1-s)}, \;\; P''(s) = \lambda^2 e^{-\lambda(1-s)}.$$

One can easily find

$$E(X) = P'(1) = \lambda,$$

$$E[X(X-1)] = P''(1) = \lambda^2,$$

$$Var(X) = P''(1) - [P'(1)]^2 + P'(1) = \lambda.$$

Example 3. Consider the PGF

$$P(s) = \left(\frac{1+s}{2} \right)^n, \;\; -\infty < s < \infty.$$

Expanding the right side into a power series, we get

$$P(s) = \sum_{k=0}^{\infty} \binom{n}{k} \frac{s^k}{2^n} = \sum_{k=0}^{\infty} p_k s^k,$$

and it follows that

$$P(X = k) = p_k = \frac{\binom{n}{k}}{2^n}, \;\; k = 0, 1, 2, ...$$

Note: We observe that the PGF, being defined only for discrete integer-valued random variables, has limited utility. We next consider a generating function that is quite useful in probability and statistics.

Moment Generating Function: Let X be a random variable defined on $(\Omega, \mathcal{F}, P)$. The function

$$M_X(t) = E[e^{tX}] \tag{2}$$

is known as the *moment generating function* (MGF) of X if the expectation on the right hand side of (2) exists in some neighborhood of the origin.

Example 4. Let X is a random variable with PMF

$$f(x) = \begin{cases} \frac{6}{\pi^2 k^2}, & k = 1, 2, ..., \\ 0, & \text{otherwise.} \end{cases}$$

Then $\frac{1}{\pi^2} \sum_{k=1}^{\infty} \frac{e^{tk}}{k^2}$, is infinite for every $t > 0$. We see that the MGF of X does not exist. In fact, $E(X) = \infty$.

Example 5. Let X is a random variable with PDF

$$f(x) = \begin{cases} \frac{1}{2} e^{-x/2}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$M_X(t) = \frac{1}{2} \int_0^{\infty} e^{(t-1/2)x} dx = \frac{1}{1-2t}, \ \ t < 1/2.$$

Example 6. Let X have the PMF

$$P(X = k) = \begin{cases} \frac{e^{-\lambda} \lambda^k}{k!}, & k = 0, 1, 2, ..., \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$M_X(t) = E[e^{tX}] = e^{-\lambda} \sum_{k=0}^{\infty} e^{tk} \frac{\lambda^k}{k!} = e^{-\lambda(1-e^t)} \ \forall \ t.$$

Theorem 1. The MGF uniquely determines a distribution function and, conversely, if the MGF exists, it is unique.

Theorem 2. If the MGF $M_X(t)$ of a random variable X exists for t in $(-t_0, t_0)$, say, $t_0 > 0$, the derivatives of all order exist at $t = 0$ and can be evaluated under the integral sign, i.e.,

$$M^{(k)}(t)|_{t=0} = E(X^k) \tag{3}$$

for positive integral k.

Remark 3. If the MGF $M_X(t)$ of a random variable X exists for t in $(-t_0, t_0)$, say, $t_0 > 0$, one can express $M_X(t)$ uniquely in a Maclaurian series expansion:

$$M_X(t) = M(0) + \frac{M'(0)}{1!}t + \frac{M''(0)}{2!}t^2 + ..., \tag{4}$$

so that $E(X^k)$ is the coefficient of $t^k/k!$ is expansion (4).

Example 7. Let X be a random variable with PDF $f(x) = \frac{1}{2}e^{-x/2},\ \ x > 0$. The MGF of X (see Example 5) is given by

$$M_X(t) = \frac{1}{1-2t},\ \ t < 1/2.$$

Thus

$$M'(t) = \frac{2}{(1-2t)^2}\ \text{ and }\ M''(t) = \frac{8}{(1-2t)^3},\ \ t < 1/2.$$

It follows that

$$E(X) = M'(t)|_{t=0} = 2,\ \ E(X^2) = M''(t)|_{t=0} = 8,\ \text{ and }\ Var(X) = 4.$$

Example 8. Let X be a random variable with PDF

$$f(x) = \begin{cases} 1, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$M_X(t) = E[e^{tX}] = \int_0^1 e^{tx}dx = \frac{e^t - 1}{t},\ \ \forall\ t,$$

$$M'_X(t) = \frac{te^t - (e^t - 1)}{t^2},$$

$$E(X) = M'(0) = \lim_{t\to 0}\frac{te^t - e^t + 1}{t^2} = \frac{1}{2}.$$

Note. We emphasize that the expectation $E[e^{tX}]$ does not exist unless t is carefully restricted. In fact, the requirement that $M_X(t)$ exists in neighborhood of zero is a very strong requirement that is not satisfied by some common distributions. We will next consider a generating function that exists for all distributions.